Artificial Neural Networks and Deep Learning

Assignment 4

Guidelines

- Show all steps clearly by hand.
- Write formulas before calculations.
- Draw diagrams where needed.
- Keep answers neat and structured.

Tokenization, Word Embeddings, Attention & Transformers

Q1. (Conceptual – Need for Tokenization)

Why can neural networks not directly process raw text?

Explain the purpose of:

- Tokenization
- Vocabulary
- Token IDs

Also explain difference between:

- Character-level tokenization
- Word-level tokenization

Q2. (Numerical – Tokenization Example)

Given the sentence:

“Deep learning models learn patterns”

Suppose the vocabulary is:

Word	ID
Deep	1
learning	2
models	3
learn	4
patterns	5

a) Convert sentence into token IDs
b) If embedding dimension is 4, what will be shape of embedding matrix for this sentence?
c) Why are embeddings preferred over one-hot vectors?

Q3. (Conceptual – Word Embeddings)

What are word embeddings?

Explain how embeddings capture:

- Semantic similarity
- Contextual meaning

Why are embeddings better than one-hot encoding?

Q4. (Conceptual – Context Problem in NLP)

Traditional embeddings assign one vector per word.

Why is this problematic for words with multiple meanings?
Use an example like:

- "bank" (river bank vs financial bank)

How do contextual embeddings improve this?

Q5. (Conceptual – Attention Mechanism)

Explain the intuition behind the **attention mechanism**.

Why was attention introduced in sequence models?
How does attention help models focus on important words?

Q6. (Numerical – Attention Score Calculation)

Suppose:

$$Q = [1,2], \quad K = [2,1]$$

Attention score is computed using dot product:

$$Q \cdot K$$

a) Compute attention score
b) What does a larger score represent?
c) Why are query and key vectors used?

Q7. (Conceptual – Self-Attention)

Explain **self-attention**.

How does a word attend to other words in the same sentence?
Why is self-attention more powerful than standard RNN processing for long sentences?

Q8. (Conceptual – Transformer Architecture)

Draw and explain the basic Transformer architecture.

Your answer should include:

- Encoder
- Decoder
- Multi-head attention
- Feedforward layers
- Positional encoding

Explain role of each component briefly.

Q9. (Conceptual – Positional Encoding)

Transformers process all words in parallel.

Why do Transformers need positional encoding?
What problem occurs without positional information?

Q10. (Conceptual + Application – Modern NLP Models)

Large language models like ChatGPT are based on Transformers.

Explain why Transformers became more successful than RNNs in NLP tasks.

Discuss:

- Parallel processing
- Long-range dependencies
- Scalability
- Training efficiency

Q11. (Conceptual – One-Hot Encoding vs Embeddings)

Compare:

- One-hot encoding
- Word embeddings

in terms of:

- Vector size
- Semantic meaning
- Sparsity
- Memory efficiency

Why do modern NLP systems prefer embeddings?

Q12. (Numerical – Embedding Matrix Dimensions)

Suppose:

- Vocabulary size = 20,000
- Embedding dimension = 300

a) What is the size of the embedding matrix?
b) How many trainable parameters does this layer contain?
c) If sentence length is 10 words, what is shape of embedded output?

Q13. (Conceptual – Multi-Head Attention)

What is **multi-head attention**?

Why does Transformer use multiple attention heads instead of only one?
Explain how different heads can learn different relationships in a sentence.

Q14. (Conceptual – Encoder vs Decoder)

Differentiate between Transformer:

- Encoder
- Decoder

Discuss their roles in:

- Language understanding
- Text generation

Give one application example for each.

Q15. (Application-Based – Transformer Use Cases)

Explain how Transformers are used in any three of the following applications:

- Machine translation
- Chatbots
- Text summarization
- Question answering
- Speech recognition
- Code generation

Why are Transformers suitable for these tasks?